

MAIA

Motion Assistance for Impaired Animals
Wearable ADAS System for Blind Dogs

Parameter	Value
Prototype Platform	XIAO ESP32S3 Plus
Production Target	ESP32-S3-WROOM-1U-N16R8
Framework	ESP-IDF v6.0 + FreeRTOS
Battery	26650 Li-Ion (5000mAh) - Removable
Target Runtime	24+ hours (camera off)
Main Unit Size	70 x 30 x 30 mm
Motor Sleeve	Adjustable, fits any collar
Version	1.5
Date	April 2026

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1. Changelog from v1.4

This revision consolidates hardware changes made during design review and production planning. The main areas affected are MCU target, IMU selection, and the addition of an object-recognition camera subsystem.

1.1 MCU Target

Production target is now the bare module **ESP32-S3-WROOM-1U-N16R8** (16MB flash, 8MB PSRAM, U.FL antenna). The XIAO ESP32S3 Plus remains the prototype board. GPIO assignments are identical on both platforms, so firmware does not change on migration.

1.2 IMU Replacement

The MPU6050 (obsoleted by TDK InvenSense) has been replaced by the **ST LSM6DSOX**. The new IMU adds a Machine Learning Core (MLC) and 16 Finite State Machines, enabling on-chip activity classification (walking, running, stationary, etc.) without CPU load on the ESP32. Power consumption drops from ~3.9 mA (MPU6050) to 0.55 mA (LSM6DSOX, combo high-perf).

1.3 Pinout Changes

Signal	Old GPIO	New GPIO	Reason
VL53L5CX #1 INT	40	42	GPIO40 reassigned to SCCB_SDA (camera)
Button extra	-	43	Console via USB CDC frees GPIO43/44
LED flash	-	44	Illumination for camera capture
CAM_PWDN	-	45	Software power control for camera
Camera DVP	-	10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031,1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047,1048,1049,1050,1051,1052,1053,1054,1055,1056,1057,1058,1059,1060,1061,1062,1063,1064,1065,1066,1067,1068,1069,1070,1071,1072,1073,1074,1075,1076,1077,1078,1079,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1090,1091,1092,1093,1094,1095,1096,1097,1098,1099,1100,1101,1102,1103,1104,1105,1106,1107,1108,1109,1110,1111,1112,1113,1114,1115,1116,1117,1118,1119,1120,1121,1122,1123,1124,1125,1126,1127,1128,1129,1130,1131,1132,1133,1134,1135,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1152,1153,1154,1155,1156,1157,1158,1159,1160,1161,1162,1163,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1174,1175,1176,1177,1178,1179,1180,1181,1182,1183,1184,1185,1186,1187,1188,1189,1190,1191,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,1217,1218,1219,1220,1221,1222,1223,1224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,2225,2226,2227,2228,2229,2230,2231,2232,2233,2234,2235,2236,2237,2238,2239,2240,2241,2242,2243,2244,2245,2246,2247,2248,2249,2250,2251,2252,2253,2254,2255,2256,2257,2258,2259,2260,2261,2262,2263,2264,2265,2266,2267,2268,2269,2270,2271,2272,2273,2274,2275,2276,2277,2278,2279,2280,2281,2282,2283,2284,2285,2286,2287,2288,2289,2290,2291,2292,2293,2294,2295,2296,2297,2298,2299,2300,2301,2302,2303,2304,2305,2306,2307,2308,2309,2310,2311,2312,2313,2314,2315,2316,2317,2318,2319,2320,2321,2322,2323,2324,2325,2326,2327,2328,2329,2330,2331,2332,2333,2334,2335,2336,2337,2338,2339,2340,2341,2342,2343,2344,2345,2346,2347,2348,2349,2350,2351,2352,2353,2354,2355,2356,2357,2358,2359,2360,2361,2362,2363,2364,2365,2366,2367,2368,2369,2370,2371,2372,2373,2374,2375,2376,2377,2378,2379,2380,2381,2382,2383,2384,2385,2386,2387,2388,2389,2390,2391,2392,2393,2394,2395,2396,2397,2398,2399,2400,2401,2402,2403,2404,2405,2406,2407,2408,2409,2410,2411,2412,2413,2414,2415,2416,2417,2418,2419,2420,2421,2422,2423,2424,2425,2426,2427,2428,2429,2430,2431,2432,2433,2434,2435,2436,2437,2438,2439,2440,2441,2442,2443,2444,2445,2446,2447,2448,2449,2450,2451,2452,2453,2454,2455,2456,2457,2458,2459,2460,2461,2462,2463,2464,2465,2466,2467,2468,2469,2470,2471,2472,2473,2474,2475,2476,2477,2478,2479,2480,2481,2482,2483,2484,2485,2486,2487,2488,2489,2490,2491,2492,2493,2494,2495,2496,2497,2498,2499,2500,2501,2502,2503,2504,2505,2506,2507,2508,2509,2510,2511,2512,2513,2514,2515,2516,2517,2518,2519,2520,2521,2522,2523,2524,2525,2526,2527,2528,2529,2530,2531,2532,2533,2534,2535,2536,2537,2538,2539,2540,2541,2542,2543,2544,2545,2546,2547,2548,2549,2550,2551,2552,2553,2554,2555,2556,2557,2558,2559	
Camera SCCB	-	39 (SCL) / 40 (SDA)	Separate I2C bus (avoids 0x3C collision with SSD1306)

2. Project Overview

MAIA (Motion Assistance for Impaired Animals) is a wearable assistance device designed for blind dogs. The system uses Time-of-Flight (ToF) sensors to detect obstacles in three directions (left, center, right) and provides haptic feedback through vibration motors on the dog's collar. An on-demand camera subsystem identifies household landmarks (feeding station, bed, doorways) to deliver contextual haptic patterns.

The system consists of two main components: the Main Unit (electronics, battery, sensors, display, camera) and the Motor Sleeve (fabric wrap with embedded vibration motors fitting any collar).

2.1 Key Features

Feature	Description
3-Zone ToF Detection	Left, Center, Right up to 4m
Object Recognition	OV3660 camera, on-demand activation
Haptic Feedback	1x DRV2605L (center) + 2x MOSFET ERM
Auto Sleep/Wake	Motion-triggered via LSM6DSOX MLC
Activity Classification	On-chip ML (walking/stationary/etc)
Temperature Monitor	Ambient monitoring with alerts
OLED Display	128x32 blue pixels for status
WiFi Sync	Cloud upload every 10 min
24h+ Battery	Removable 26650 for nightly charging
Universal Mount	Sleeve fits any collar via velcro

3. System Architecture

3.1 Block Diagram

Block	Interface	Function
ESP32-S3-WROOM-1U-N16R8	-	Main controller, WiFi, BLE
VL53L5CX #1	I2C (0x54)	Left/Center ToF sensor
VL53L5CX #2	I2C (0x52)	Right/Center ToF sensor
LSM6DSOX	I2C (0x6A)	Motion/activity, MLC-based sleep/wake
SSD1306	I2C (0x3C)	OLED display 128x32
DS18B20	1-Wire	Temperature sensor
DRV2605L	I2C (0x5A)	Haptic driver center motor
OV3660	DVP + SCCB (0x3C)	3MP camera via FPC
Motor Left	PWM + MOSFET	ERM via 2N7000 - GPIO7
Motor Right	PWM + MOSFET	ERM via 2N7000 - GPIO8
LED Flash	PWM + MOSFET	Camera illumination - GPIO44
Button Main	GPIO9	User interface input
Button Extra	GPIO43	Secondary input
26650 Battery	BAT+/BAT-	5000mAh removable

3.2 I2C Bus Addresses

MAIA uses two separate I2C buses to avoid the 0x3C address collision between SSD1306 and OV3660:

Bus	Device	Address	Notes
Main (GPIO5/6)	SSD1306	0x3C	OLED Display
Main	VL53L5CX #2	0x52	ToF Right (default boot)
Main	VL53L5CX #1	0x54	ToF Left (reassigned at init)
Main	DRV2605L	0x5A	Haptic driver
Main	LSM6DSOX	0x6A	IMU
SCCB (GPIO39/40)	OV3660	0x3C	Isolated, camera-only

4. Hardware Design

4.1 Haptic Motor Configuration

Motor	Position	Driver	Interface
Center	Front (chin)	DRV2605L	I2C 0x5A
Left	Left side (ear)	2N7000 MOSFET	PWM GPIO7
Right	Right side (ear)	2N7000 MOSFET	PWM GPIO8

4.2 Sensor Configuration

Sensor	Model	Function
ToF #1	VL53L5CX	Left/Center obstacle, 8x8 zones
ToF #2	VL53L5CX	Right/Center obstacle, 8x8 zones
IMU	LSM6DSOX	6-axis + MLC activity recognition
Temperature	DS18B20	Ambient temp, emergency alerts
Camera	OV3660	3MP DVP, on-demand object recognition
Display	SSD1306	128x32 OLED, blue pixels

5. Motor Driver Circuit

5.1 Component Description

Component	Value	Function
R1 (series)	100R	Limits gate current. Protects GPIO. Max 33mA.
R2 (pull-down)	10k	Ensures motor OFF during ESP32 boot (Hi-Z).
MOSFET	2N7000	N-Ch logic-level. Switches motor GND. Vth ~2V.
Flyback Diode	SL54F	Anti-parallel with motor. Protects spike.

Total for 2 MOSFET-driven motors: 2x 2N7000, 2x 100R, 2x 10k, 2x SL54F.

The same circuit topology is reused for the camera LED flash driver on GPIO44.

6. Pinout Configuration

Pinout valid for both XIAO ESP32S3 Plus (prototype) and ESP32-S3-WROOM-1U-N16R8 (production). Console via USB CDC (GPIO19/20) - UART0 pins (GPIO43/44) are available for general-purpose I/O.

6.1 Core Peripherals

GPIO	Function	XIAO Plus	Pull-up
1	VL53L5CX #1 LPn	D0	Ext 47k
2	VL53L5CX #2 LPn	D1	Ext 47k
3	DS18B20 DATA	D2	Ext 4.7k
4	LSM6DSOX INT1	D3	Internal
5	I2C SDA (main)	D4	Ext 2.2k
6	I2C SCL (main)	D5	Ext 2.2k
7	Motor Left PWM	D8	-
8	Motor Right PWM	D9	-
9	Button main	D10	Internal
41	VL53L5CX #2 INT	D14	Internal
42	VL53L5CX #1 INT	D15	Internal
43	Button extra	D6	Internal
44	LED flash PWM	D7	-
45	CAM_PWDN	-	Ext 10k PD

6.2 Camera Interface (DVP + SCCB)

GPIO	Camera Signal	Direction
10	XCLK	Out (LEDC, 20MHz)
11	Y8	In
12	Y7	In
13	PCLK	In
14	Y6	In
15	Y2	In
16	Y5	In
17	Y3	In
18	Y4	In
38	VSYNC	In
39	SCCB_SCL	Out (open-drain)
40	SCCB_SDA	Bidir (open-drain)
47	HREF	In
48	Y9	In

6.3 Reserved / Do Not Use

GPIO	Reason
0	Boot strap (internal pull-up)
19, 20	USB D-/D+ (native USB CDC)
35, 36, 37	Octal PSRAM (N16R8 internal)
46	Strapping pin (ROM msg enable)

7. Camera Subsystem

The OV3660 3MP camera is used for on-demand object recognition. It is powered down by default (PWDN=HIGH) to preserve battery. ToF triggers camera activation when a stable object is detected at medium range.

7.1 Connector

FPC 24-pin, 0.5mm pitch, bottom contact, right angle SMD, flip-lock actuator. Compatible with standard ESP32-CAM OV3660/OV2640/OV5640 modules.

Parameter	Value
Manufacturer	Bossie
Part	BX-FPC05-1.5H24PX
LCSC	C18078354
Pitch	0.5mm
Contacts	24 + 2 shield pads
Current Rating	500mA
FPC Thickness	0.3mm

7.2 FPC Pinout

FPC Pin	Signal	ESP32 GPIO / Net
1	VCC (3V3)	3v3_esp
2	GND	GND
3	SCCB_SCL	GPIO39
4	SCCB_SDA	GPIO40
5	VSYNC	GPIO38
6	HREF	GPIO47
7	PCLK	GPIO13
8	XCLK	GPIO10
9	Y9	GPIO48
10	Y8	GPIO11
11	Y7	GPIO12
12	Y6	GPIO14
13	Y5	GPIO16
14	Y4	GPIO18
15	Y3	GPIO17
16	Y2	GPIO15
17	GND	GND
18	RESET	Pull-up 10k to 3V3
19	PWDN	GPIO45
20-24	GND	GND
25, 26	Shield (mechanical)	GND

7.3 External Components

Location	Component	Value	Purpose
Pin 1 (VCC)	C14	4.7uF MLCC	Bulk decoupling
Pin 1 (VCC)	C15	100nF MLCC	HF decoupling
Pin 3 (SCL)	R16	3.3k to 3V3	SCCB pull-up
Pin 4 (SDA)	R17	3.3k to 3V3	SCCB pull-up
Pin 18 (RESET)	R15	10k to 3V3	Release
Pin 19 (PWDN)	R14	10k to GND	Default power-down

7.4 Power Budget

OV3660 active: ~150 mA @ 3.3V. Powered from 3v3_esp rail (AP7365-33 LDO, 600mA max). Peak rail demand with camera + ESP32 WiFi TX + ToF = ~500mA, within LDO limits. Camera duty cycle is low in normal operation (triggered only by ToF).

7.5 Rail Voltages

The OV3660 module integrates internal regulators for AVDD (2.8V analog), DOVDD (1.8V digital I/O), and DVDD (1.5V core). A single 3.3V rail on FPC pin 1 is all that is required externally.

8. IMU: LSM6DSOX

The LSM6DSOX replaces the obsolete MPU6050 from v1.4. It is a 6-axis IMU from STMicroelectronics with significant advantages for wearable always-on applications.

8.1 Specifications

Parameter	Value
Package	LGA-14L, 2.5 x 3.0 x 0.83 mm
Accelerometer range	+/- 2/4/8/16 g
Gyroscope range	+/- 125/250/500/1000/2000 dps
ODR range	1.6 Hz to 6.7 kHz
Supply VDD	1.71 V to 3.6 V
Current, combo HP	0.55 mA typ
Current, sleep	~3 uA
Interfaces	I2C, SPI, MIPI I3C
FIFO	9 KB
I2C Address	0x6A (SA0=0) / 0x6B (SA0=1)
Operating temp	-40 to +85 C

8.2 On-Chip Smart Features

The LSM6DSOX enables significant ESP32 CPU offload:

Feature	Benefit for MAIA
Machine Learning Core (MLC)	Classify dog activity (walk/run/idle/sleep) on-chip
16 Finite State Machines	Detect custom motion patterns (shaking, scratching)
Pedometer (HW)	Step counter without CPU
Wake-on-motion	ESP32 deep sleep until dog moves
Free-fall detection	Emergency event (collar dropped)
6D orientation	Head position tracking
Tap / double-tap	Owner interaction (tap device for status)

8.3 Development Flow for MLC

1. Collect labeled IMU data (walk, run, idle, sleep, etc.) using the LSM6DSOX eval board. 2. Train decision tree in ST Edge AI Suite (free tool). 3. Export configuration as .ucf register dump. 4. Firmware loads .ucf at boot. 5. Chip outputs activity class via register read. ESP32 CPU load for classification: effectively zero.

9. Power Management

9.1 Architecture

Power input from either USB-C (VBUS) or 26650 Li-Ion battery (VBAT). The TPS2116 power mux selects the active source automatically, with priority on VBUS when available.

Block	Part	Function
Power Mux	TPS2116DRL	VBUS/VBAT auto-select
Charger	BQ24040DSQR	Li-Ion charge, 26650 compatible
LDO #1	AP7365-33WG-7	3v3_esp rail (600mA)
LDO #2	AP7365-33WG-7	3v3_haptic rail (isolated)
TVS	SMAJ5.0CA (or SMF5.0CA)	VUSB surge protection
USB ESD	USBLC6-2SC6	D+/D- ESD clamp
Battery Gauge	XB5352A	VBAT voltage monitor

9.2 Current Budget

Component	Active (mA)	Sleep (uA)
ESP32-S3 idle	20	-
ESP32-S3 WiFi TX peak	240	-
ESP32-S3 deep sleep	-	14
VL53L5CX x 2 (ranging)	80	6
LSM6DSOX (combo HP)	0.55	3
SSD1306 active	20	10
DS18B20	1	1
DRV2605L idle	2	2
OV3660 active	150	0
OV3660 power-down	0	200
Peak with camera + WiFi	~500	-
Typical active (no cam, no TX)	~50	-
Deep sleep total	-	~30

At ~50 mA typical active, the 5000mAh battery yields over 100 hours of continuous use. Aggressive power management with LSM6DSOX wake-on-motion pushes real runtime beyond the 24h target.

10. USB Subsystem

The ESP32-S3 native USB (GPIO19/20) is used for both console (CDC) and flashing. No external USB-to-UART bridge is required.

Element	Part / Value	Purpose
Connector	USB-C 16-pin 2MD	Host / charge
CC pull-downs	5.1k x 2	USB-C Rd (sink)
VBUS bulk cap	4.7uF + 1uF + 100nF	Input decoupling
Ferrite bead	600R @ 100MHz	VBUS noise filter
TVS (VBUS)	SMAJ5.0CA (or SMF5.0CA)	Surge / ESD clamp
ESD array (D+/D-)	USBLC6-2SC6	Line ESD protection
Series termination	22R x 2 (D+ and D-)	Source impedance match
Diff impedance	90 ohm diff (PCB)	USB 2.0 Full-Speed

The 22 ohm series resistors are placed near the ESP32 (source termination). The USBLC6-2SC6 is placed near the USB-C connector (ESD source). Signal path: USB-C to USBLC6 to 22 ohm to ESP32.

11. Mechanical Design

11.1 Main Unit Dimensions

Dimension	Value	Notes
Length	70mm	Along collar
Width	30mm	Perpendicular
Height	30mm	Vertical
Weight	~120g	With battery
Material	ABS/PETG	IP54 rated

11.2 Battery Removal

1. Press the red latch button. 2. Slide battery compartment outward. 3. Remove 26650 cell for charging.

12. Motor Sleeve System

12.1 JST Cable Pinout

Pin	Signal	Color (suggested)
1	Motor Left (+)	Yellow
2	Motor Center (+)	Orange
3	Motor Right (+)	Red
4	GND (common)	Black
5	GND (common)	Black

13. Haptic Feedback System

13.1 Vibration Patterns

Effect	Pattern	Use Case
Soft Bump	50%, 100ms	Object 2-4m
Medium Buzz	70%, 150ms	Object 1-2m
Strong Alert	100%, 200ms	Object 0.5-1m
Rapid Pulse	100%, 50ms x3	Object <0.5m
Continuous	80%, sustained	Imminent collision
Landmark: food bowl	Pattern A (camera)	Recognized bowl
Landmark: dog house	Pattern B (camera)	Recognized house

13.2 Ignore Mode

Prevents desensitization during feeding/drinking. Triggers: distance stable <20cm for >3s, velocity ~0 (LSM6DSOX idle classification). Exit: distance >50cm, movement detected, or 5 min timeout.

14. Display Interface

14.1 Main Button Multi-Function

Press	Screen Content
1st	Battery icon + percentage + voltage
2nd	Temperature: current + high/low
3rd	Pet Name
4th	Owner Name + phone (scroll)
5th	Address (scroll)
6th	WiFi: last sync + SSID
7th	Activity: last MLC class + duration
8th	System uptime + version
9th+	Display OFF

14.2 Extra Button

GPIO43. Reserved for future use (panic trigger, manual WiFi sync, voice announcement).

15. Firmware Architecture

15.1 Software Layers

Layer	Modules
Application	main, tasks, state machine
Services	obstacle_detection, haptic_feedback, object_recognition, activity_tracker, power_manager, data_logger, st
Drivers	vl53l5cx, lsm6dsox, drv2605l, ssd1306, button, ds18b20, ov3660_camera
maia_board (BSP)	pinout, i2c_init, gpio_init, pwm_init, onewire_init, cam_init
ESP-IDF / FreeRTOS	Hardware abstraction layer

15.2 FreeRTOS Tasks

Task	Priority	Stack	Function
task_sensing	10	8KB	ToF + LSM6DSOX reading
task_haptic	15	2KB	Motor control
task_camera	8	16KB	Object recognition (on-demand)
task_display	3	4KB	OLED + button
task_monitor	1	4KB	Temp + WiFi sync + activity log

16. VL53L5CX Driver API (MAIA)

The VL53L5CX MAIA driver exposes a clean, Kconfig-driven API for dual ToF sensor management. All public functions use the **maia_tof_*** prefix. Defined in components/drivers/vl53l5cx/include/vl53l5cx.h - Copyright 2026 Vinicius May, Apache 2.0.

16.1 Constants

Constant	Value	Description
MAIA_TOF_SENSOR_LEFT	0	Left sensor ID
MAIA_TOF_SENSOR_RIGHT	1	Right sensor ID
MAIA_TOF_SENSOR_MAX	2	Total sensors
MAIA_TOF_POWER_ACTIVE	0	Full operation
MAIA_TOF_POWER_LOW_POWER	1	Sleep (retains FW)
MAIA_TOF_MAX_ZONES	64	8x8 resolution

16.2 Data Type: maia_tof_data_t

Ranging result for one sensor. Up to 64 zones (8x8). Zones filtered by Kconfig (status 5 or 6) return distance_mm = -1.

16.3 Public Functions

Function	Description
maia_tof_init()	Initialize both sensors. LPn-based address reassignment (#1: 0x52 to 0x54; #2 stays 0x52). U
maia_tof_deinit()	Stop ranging, remove I2C device handles, disable sensors via LPn.
maia_tof_start_ranging()	Start ranging on all initialized sensors. Mode set by Kconfig at init.
maia_tof_stop_ranging()	Stop ranging on all initialized sensors.
maia_tof_data_ready(id, *ready)	Poll specified sensor for new data.
maia_tof_get_data(id, *data)	Read latest ranging frame. Applies status filters per Kconfig.
maia_tof_set_power_mode(id, mode)	Set power mode. LOW_POWER retains firmware in RAM (no re-upload on wake).

17. Bill of Materials

17.1 Main Unit Electronics

Item	Part Number	Qty	Notes
MCU Module	ESP32-S3-WROOM-1U-N16R8	1	Espressif, 16MB flash / 8MB PSRAM
Prototype Board	XIAO ESP32S3 Plus	1	Seeed Studio (dev only)
ToF Sensor	VL53L5CX	2	ST, LCSC C3178303
IMU	LSM6DSOX	1	ST, 6-axis + MLC
Haptic Driver	DRV2605L	1	TI, center motor
OLED Display	SSD1306 128x32	1	Blue, I2C 0x3C
Temp Sensor	DS18B20+	1	TO-92
Camera Module	OV3660 FPC 24-pin	1	3MP, on-demand
FPC Connector	Bossie BX-FPC05-1.5H24PX	1	LCSC C18078354
MOSFET	2N7000	3	Motor L, Motor R, LED flash
Flyback Diode	SL54F	2	Motor protection
Battery	26650 Li-Ion	1	5000mAh
Button	Tactile 6x6mm	2	Main + Extra
JST Connector	JST-PH 5-pin	1	Motor cable

17.2 Power Section

Item	Part Number	Qty
Power Mux	TPS2116DRL	1
Charger	BQ24040DSQR	1
LDO 3.3V	AP7365-33WG-7	2
Battery Monitor	XB5352A	1
TVS VBUS	SMAJ5.0CA (or SMF5.0CA)	1
USB ESD	USBLC6-2SC6	1
Ferrite Bead	600R @ 100MHz, 0603	1
USB Connector	USB-C 16-pin 2MD	1

17.3 Resistors

Value	Qty	Purpose
22R 1%	2	USB D+/D- series term
100R	3	MOSFET gate series
2.2k	2	I2C pull-up main
3.3k	2	SCCB pull-up camera
4.7k	1	1-Wire pull-up DS18B20
5.1k	2	USB-C CC pull-down
10k	5	MOSFET PD, RESET, PWDN
47k	2	VL53L5CX LPn pull-up

17.4 Capacitors

Value	Qty	Purpose
100nF MLCC	~10	HF decoupling all rails
1uF MLCC	2	VBUS / battery monitor

Value	Qty	Purpose
4.7uF MLCC	~5	Bulk decoupling (cam, ESP32, LDO)
10uF MLCC	2	Power mux / charger

17.5 Motor Sleeve

Item	Description	Qty
Vibration Motor	ERM 10mm, 3V	3
Fabric Sleeve	Neoprene 250x40mm	1
Velcro	Hook + Loop strips	2
Flat Cable	5-wire, 150mm	1
JST Plug	JST-PH 5-pin female	1

Estimated Cost: ~\$75-90 USD per unit (excluding enclosure and PCB fabrication). Camera adds ~\$8-10 versus v1.4.